

PROJECT PROPOSAL

A. GENERAL INFORMATION

A1. Project Title

"High-Fidelity Digital Twin Reconstruction of Base Transceiver Stations: Enhancing Vanilla 3DGS for Accelerated Training and Superior Rendering"

A2. Duration

3 months .

A3. Principal Investigator

Full name: Lê Anh Kiệt

Student ID: ITCSIU24047

University: International University – Vietnam National University Ho Chi Minh City

A4. Project Members (including the Principal Investigator)

No.	Full name	Student ID	University / School
1	Lê Anh Kiệt	ITCSIU24047	International University – VNU-HCM

B. RESEARCH DESCRIPTION

B1. Introduction

3D Gaussian Splatting (3DGS) enables the reconstruction of 3D models of BTS towers directly from on-site photographs, supporting the creation of digital twins for remote monitoring and maintenance. However, the training process of vanilla 3DGS is still slow and generates many redundant Gaussians on flat surfaces (the tower body, the ground), while not focusing enough on complex structural details such as antennas, steel frames, and mechanical components on the tower. This reduces efficiency when many BTS sites need to be reconstructed quickly over a wide area.

This project surveys and verifies several 3DGS optimization techniques (structure-guided densification, smart pruning, faster optimization), then combines the most effective techniques into a single pipeline for BTS towers that trains faster and renders with higher fidelity than vanilla 3DGS.

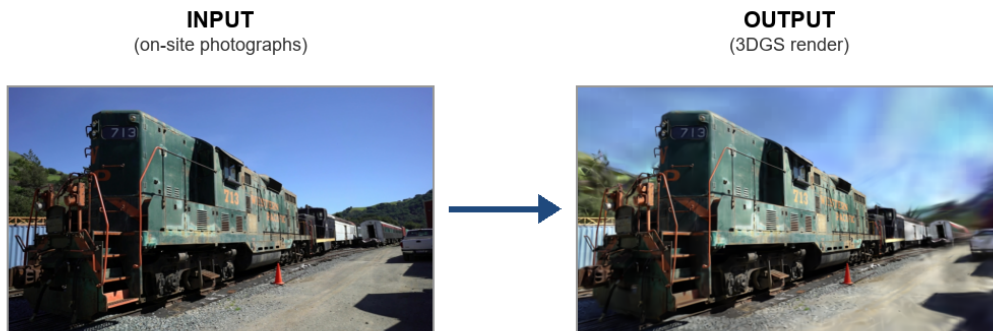


Figure 1: 3DGS pipeline overview: a set of on-site photographs (input) is used to train a set of 3D Gaussians, which are then rendered into novel-view images (output).

B2. Objectives and Research Content

- Study the original 3DGS pipeline and its training/density-control bottlenecks as the baseline.
- Survey multiple published techniques for faster training and better densification, and verify each one against its own source code.
- Select and combine the most effective techniques into one optimized pipeline for BTS digital-twin reconstruction.
- Benchmark the combined pipeline against vanilla 3DGS using PSNR, SSIM, LPIPS, Gaussian count, training time, and memory use.

B3. Expected Results

- A short, verified report and slides comparing multiple 3DGS optimization techniques.
- An optimized 3DGS pipeline for BTS digital twins, combining the best-performing techniques.
- A real benchmark table comparing the optimized pipeline with vanilla 3DGS.